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Vellore Institute of Technology

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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

LAB WEEK - 5

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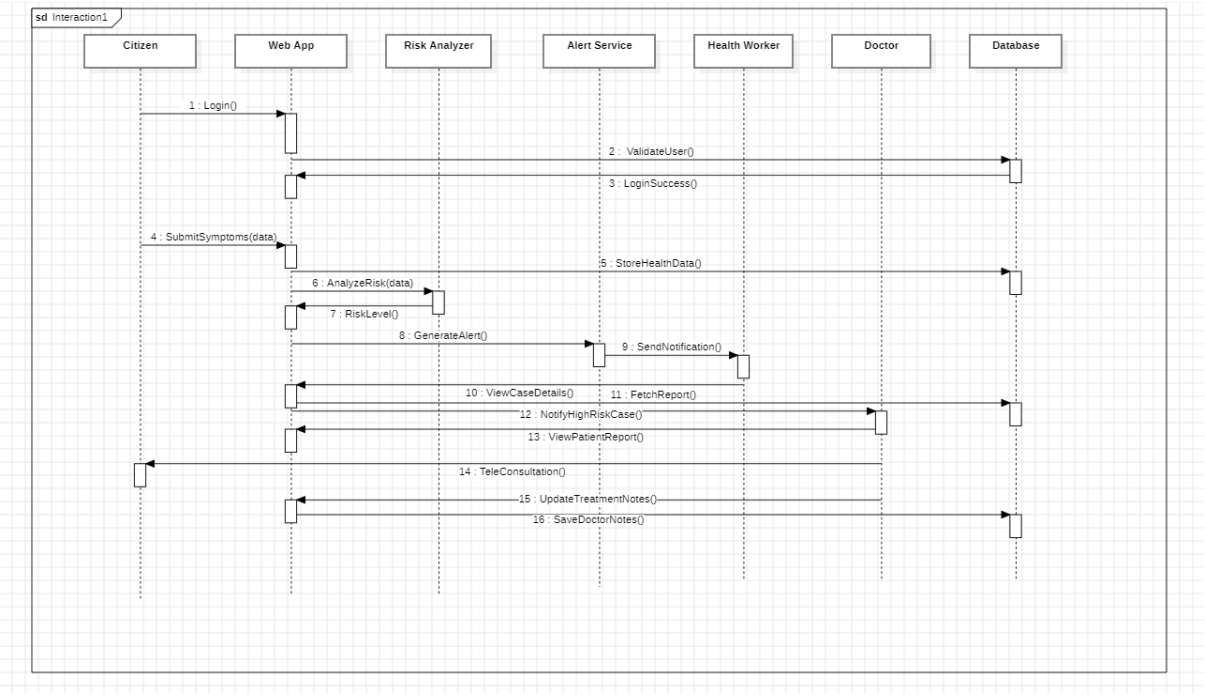
SLOT: D2 + TD2

SEQUENCE DIAGRAM



This sequence diagram illustrates the interaction flow in the **Community Micro-Health Monitoring & Alert System** from the moment a citizen logs into the system to the point where medical intervention is provided. The citizen first submits daily health symptoms through the web application, which stores the data securely in the database. The system then sends this information to the risk analyzer module, which evaluates the data using predefined rules and classifies the case as low, medium, or high risk.

Lifelines Used in the Diagram Citizen – Represents community members who log in and submit daily health symptoms.

Web App – Acts as the central system interface that receives inputs, processes requests, and coordinates with other components.

Risk Analyzer – Evaluates submitted health data using rule-based logic to determine the risk level.

Alert Service – Generates and sends notifications for medium and high-risk cases.

Health Worker – Receives alerts, reviews cases, and provides initial medical guidance.

Doctor – Handles high-risk cases by reviewing reports, conducting tele-consultations, and updating treatment plans.

Database – Stores user details, health reports, and medical records securely.

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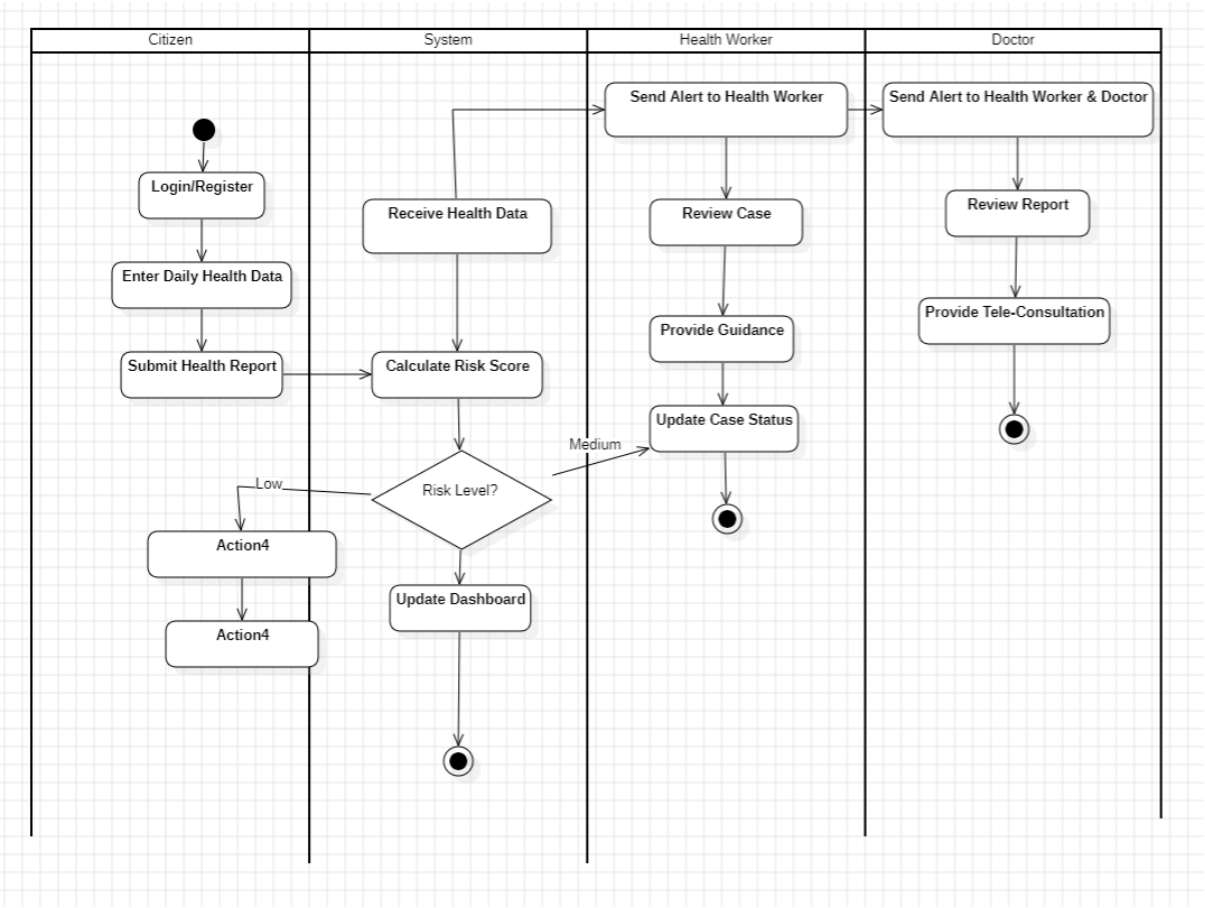
ACTIVITY DIAGRAM



The activity diagram represents the workflow of the Community Micro-Health Monitoring & Alert System from user registration to medical response. It shows how different users (citizens, system, health workers, and doctors) interact with the system and how decisions are made at each stage.Major Classes and Their Roles:

Citizen

- Registers and logs into the system
- Enters daily health symptoms and basic health parameters
- Submits health report for analysis

System

- Validates user input data
- Calculates risk level using rule-based logic
- Classifies reports as Low, Medium, or High risk
- Stores health records in the database

- Sends alerts for Medium and High-risk cases

Health Worker

- Receives alerts from the system
- Reviews citizen health reports
- Contacts citizens if necessary
- Provides basic medical guidance or instructions

Doctor

- Views detailed health reports
- Provides tele-consultation when required
- Suggests treatment or further medical actions

Administrator

- Manages users and system data
- Monitors community health trends through dashboards
- Ensures system reliability and smooth functioning